Engineering Materials
ME 282

Frank Ofori Agyemang (PhD)

PB 320

Phone No.; 0555096985

Assisted by

ELIZABETH HENEWAA AKOTO (MPhil)

EMMANUEL SEY (TA)

PRE-REQUISITE

- Basic Chemistry

Tentative Forms of Assessment

Assignments	**5**
Quizzes	**10**
Mid-Sem Exam	**15**
Final Exam	**70**
Total	**100**

Course Outline

Fundamentals

- Introduction

Classification of materials

- Metals
- Ceramics
- Polymers
- Composite
- Electronic materials

Crystal Structures

- Crystalline and non-crystalline
- Simple cubic
- Body centered cubic
- Face centered cubic

Control and Processing

- Phase Diagrams
- Phase Transformations
- Thermal processing

Mechanical Behavior

- Elastic deformation
- plastic deformation

Text

- *Materials Science and Engineering: An Introduction*, William D. Callister, John Wiley & Sons, 2010.
- *Fundamentals of Materials Science & Engineering: an integrated approach*, Callister and Rethwisch (Special 3rd edition for MSE282)
- *Materials Science and Engineering*, V. Raghavan, Fifth Edition, Prentice Hall of India Pvt. Ltd., New Delhi, 2004.

“WITHOUT MATERIALS, THERE IS NO ENGINEERING.”



Introduction

What type of materials have been used for these beverage containers?



A



C

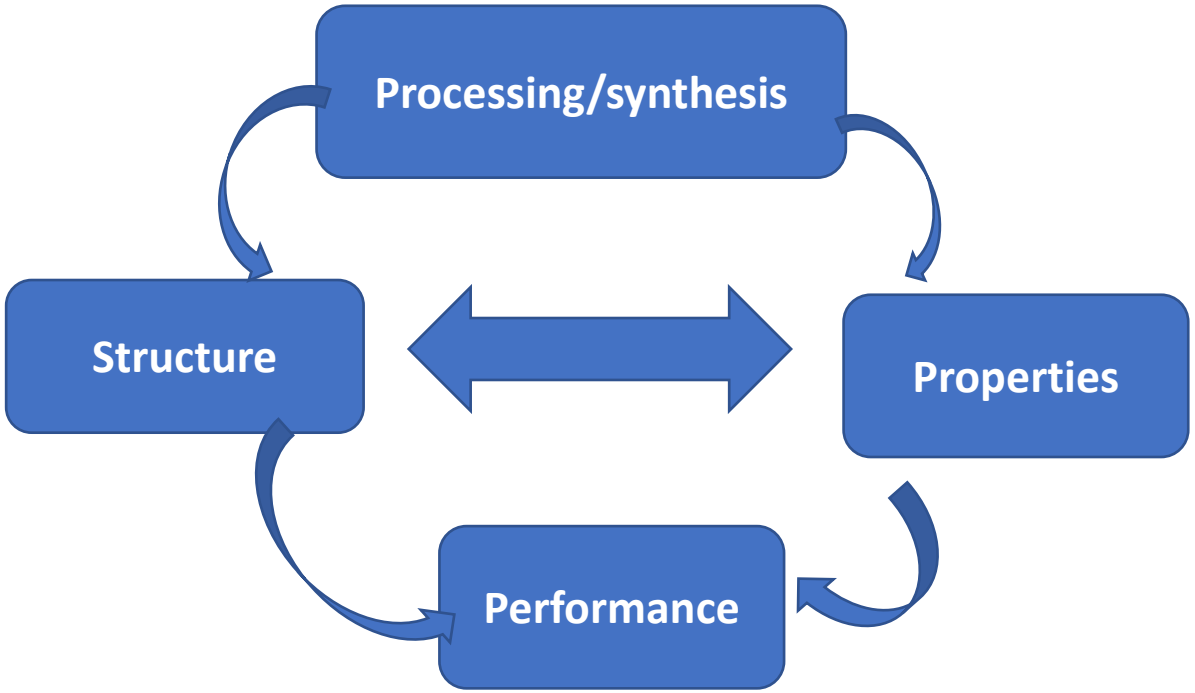


B

Introduction Cont….

➢ What is Materials Science and Engineering?

- Relationship between the four components:

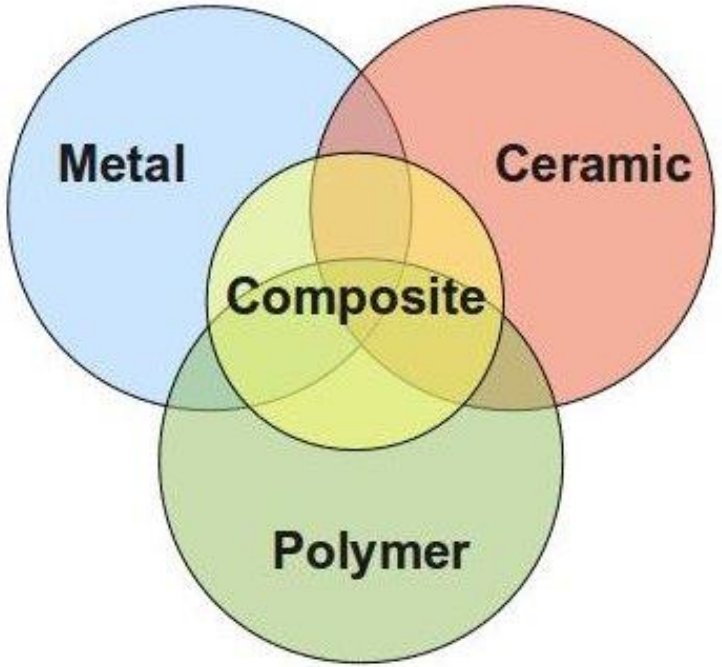


Why the Study of Materials is Important?

- **Production and processing of materials constitute a large part of developed country's economy.**
- **Engineers choose materials to suite design.**
- **New materials might be needed for some new applications.**
  - **Example :- High temperature resistant materials.**
  - **Space station and Mars Rovers should sustain conditions in space.**
  - ***High speed, low temperature, strong but light.**
- **Modification of properties might be needed for some applications.**
  - **Example :- Heat treatment to modify properties.**

Introduction Cont….

- Types of Materials

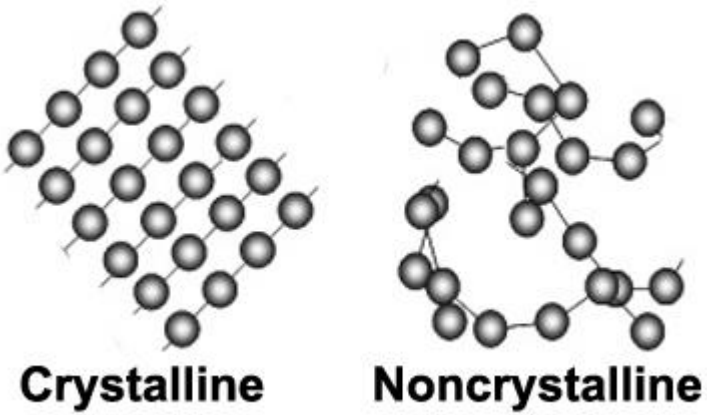


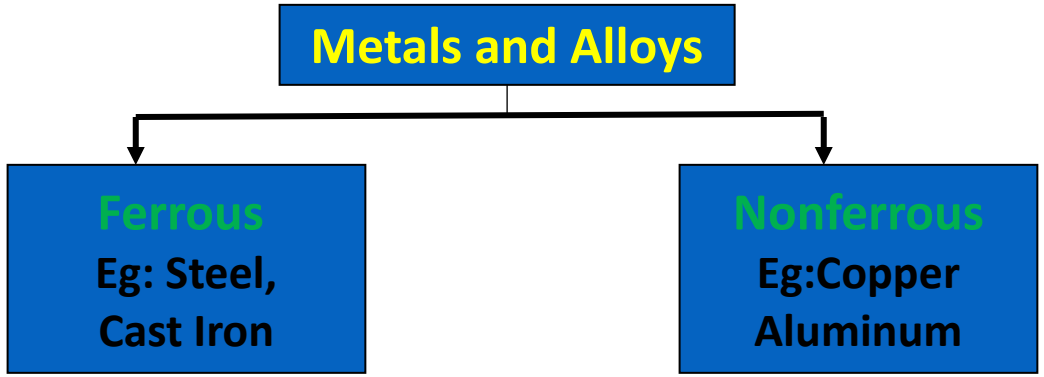
- Advanced Materials: such as Semiconductors, biomaterials, energy materials, nanomaterials, smart materials, etc. are obtained from these four main class of materials

Types of materials

❖Metals

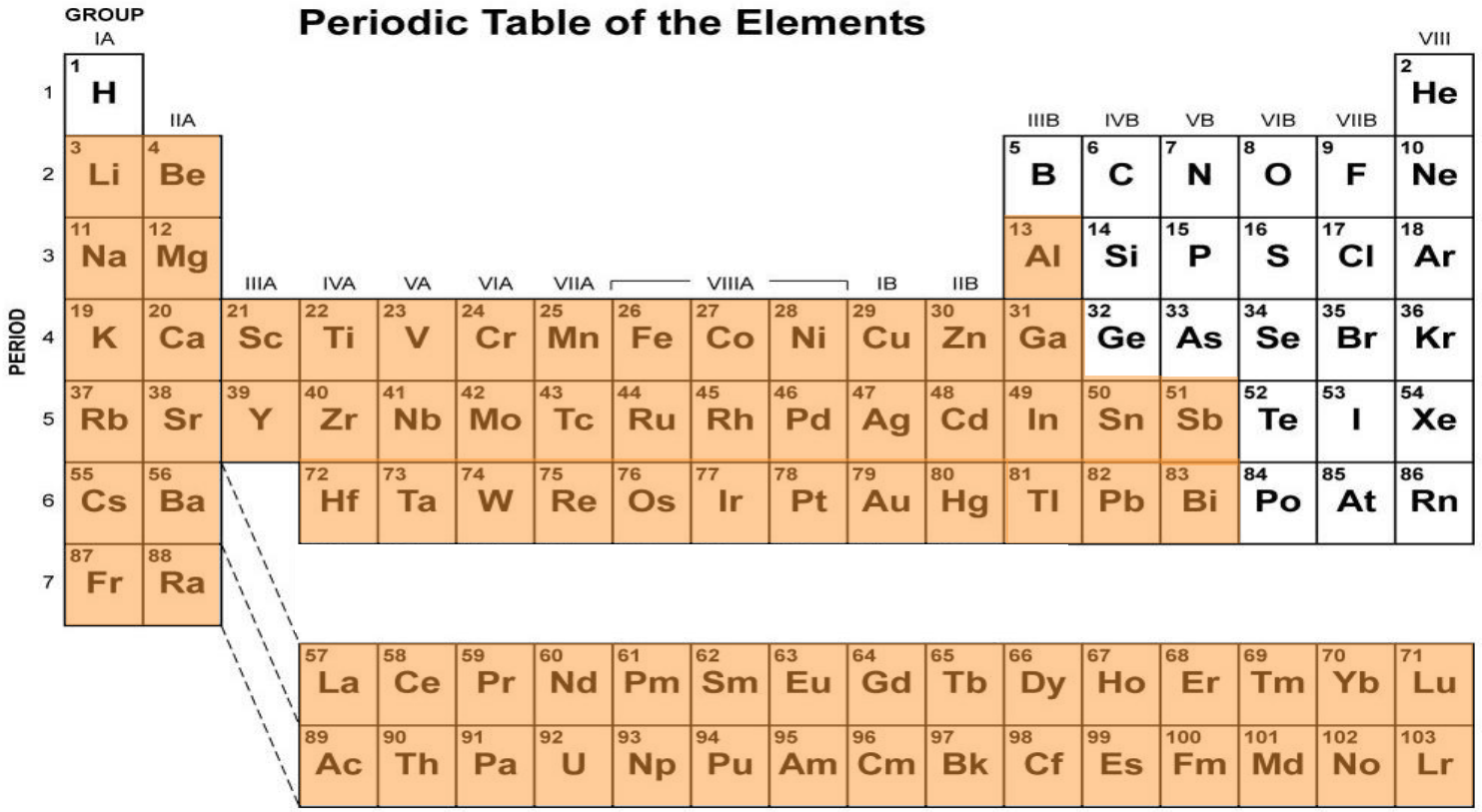
- ➢Constitute one or more metallic elements
  - ❑ *Example:- Iron, Copper, Aluminum, Lead, Zinc, etc.*
- ➢Orderly atomic arrangement
- ➢ Inorganic and have crystalline structure.
- ➢Relatively dense
- ➢ Good thermal and electric conductors





Metals

Atoms arranged in a regular repeating structure



Properties

- good conductors of electricity and heat
- lustrous appearance
- susceptible to corrosion
- strong, but deformable

Metal Applications

Applications

- Electrical wiring
- Structures: buildings, bridges, etc.
- Automobiles: body, chassis, springs, engine block, etc.
- Airplanes: engine components, fuselage, landing gear assembly, etc.
- Trains: rails, engine components, body, wheels
- Machine tools: drill bits, hammers, screwdrivers, saw blades, etc.
- Shape memory materials: eye glasses
- Magnets
- Catalysts

Examples

- Pure metal elements (Cu, Fe, Zn, Ag, etc.)
- Alloys (Cu-Sn=bronze, Cu-Zn=brass, Fe-C=steel, Pb-Sn=solder, NiTinol)
- Intermetallic compounds (e.g. Ni_3Al)

Introduction Cont….

- Some metallic products











Types of Materials

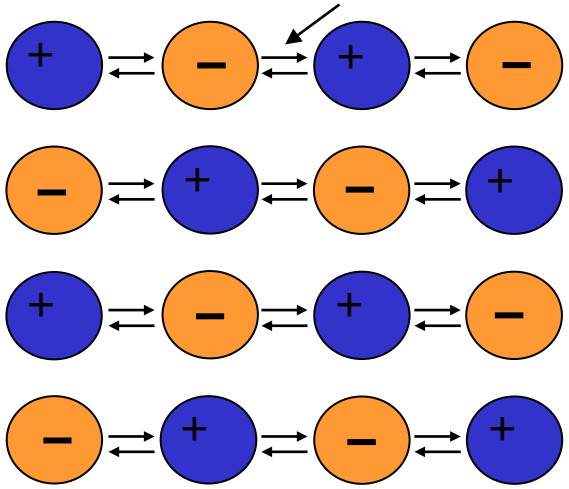
Ceramics and Glasses

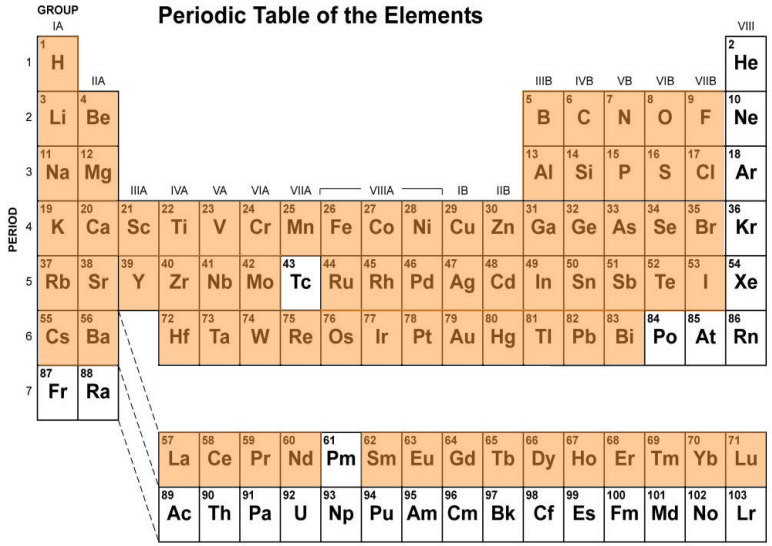
- **Metallic and nonmetallic elements are chemically bonded together.**
  - **Inorganic but can be either crystalline, noncrystalline or mixture of both.**

- **Except for glasses, atoms are regularly arranged**
- **Typically covalent/ionic. In some cases highly directional covalent bonding. Ionic in case of Al_2O_3, SiO_2 glasses and slags**

Ceramics and Glasses

Coulombic bonding force



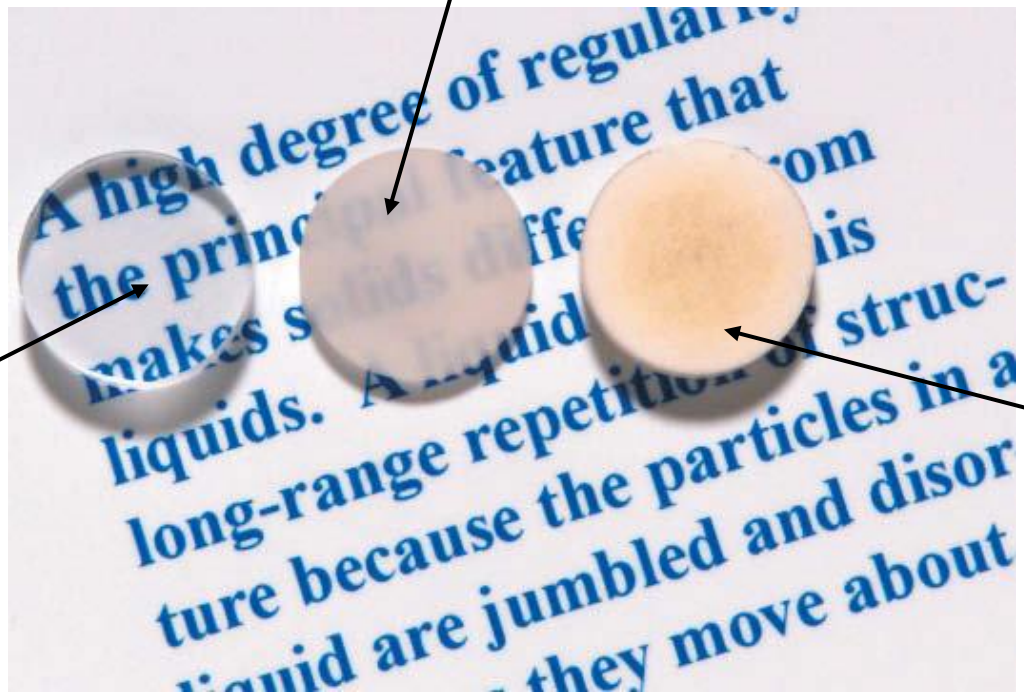


Properties

- Thermally and electrically insulating
- Stronger than metals
- Lower density than most metals
- Low resistance to fracture: low toughness or brittle
- Low ductility or malleability: low plasticity
- Resistant to high temperatures and harsh environments
- Wear resistant (hard), but brittle
- Chemical stability: corrosion resistant
- Good optical properties
- Adhesives

Introduction Cont….

- **Ceramic Materials**
  - **Optical properties**
  - Aluminum oxide can be
  - **Transparent**
  - **Translucent**
  - **Opaque**



Adapted from Fig. 1.2 Callister 9e, (Specimen preparation, P.A. Lessing; photo by J. Telford)

Ceramic Applications

Applications

- Electrical insulators
- Abrasives
- Thermal insulation and coatings
- Windows, television screens, optical fibers (glass)
- Corrosion resistant applications
- Electrical devices: capacitors, varistors, transducers, etc.
- Highways and roads (concrete)
- Biocompatible coatings (fusion to bone)
- Self-lubricating bearings
- Magnetic materials (audio/video tapes, hard disks, etc.)
- Optical wave guides
- Night-vision

Examples

- Simple oxides (SiO_2, Al_2O_3, Fe_2O_3, MgO)
- Mixed-metal oxides ($SrTiO_3$, $MgAl_2O_4$, $YBa_2Cu_3O_{7-x}$, having vacancy defects.)
- Nitrides (Si_3N_4, AlN, GaN, BN, and TiN, which are used for hard coatings.)

Introduction Cont….

- Some ceramic products



Introduction Cont….

- **Polymers**

- Include rubbers, plastics and many adhesives

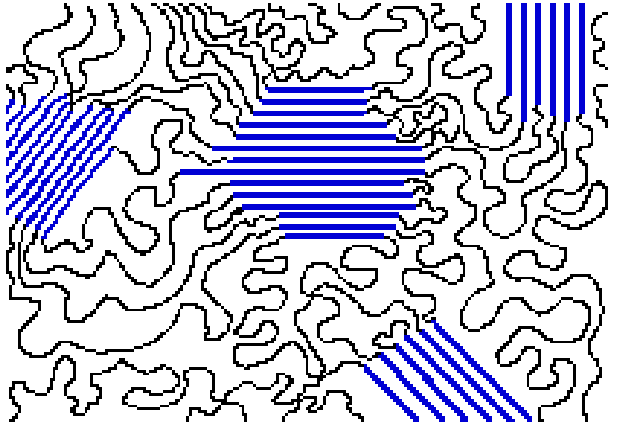
- Most of them are organic compounds based on C, H and other non metallic elements(N, Cl, Si, O etc)

- Large molecular structures, often chain-like in nature that have a backbone of C - atoms

- Examples PE, PS, PC, PVC, PP, nylon, silicone rubber etc

Introduction Cont….

- **Polymeric Materials**



Characteristics

- Mostly noncrystalline.
- Some are mixtures of crystalline and noncrystalline regions.
- Poor conductors of electricity and hence used as insulators.
- Low strength, high ductility.
- Low densities and decomposition temperatures.
- Relatively inert chemically and un-reactive in many environments
- Can be formed into complex shapes

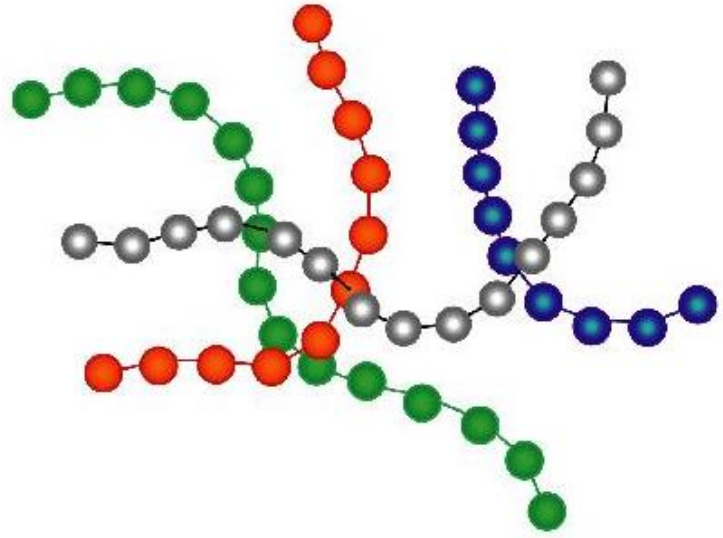
- *Examples :- Poly vinyl Chloride (PVC), Polyester, Polyethylene, Elastomers etc*

Classes and Properties: Polymers

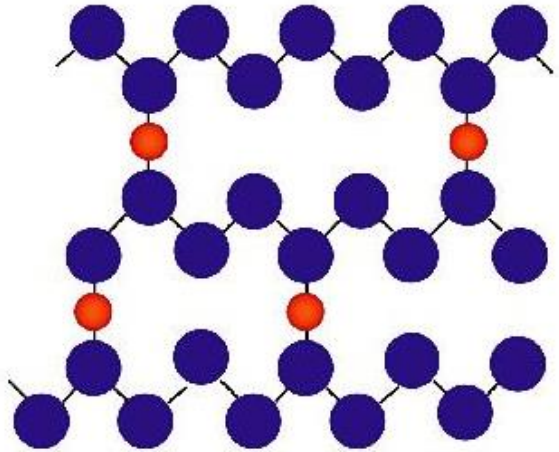
Two main types of polymers are thermosets and thermoplastics.

• Thermosets are cross-linked polymers that form 3-D networks, hence are strong and rigid.

• Thermoplastics are long-chain polymers that slide easily past one another when heated, hence, they tend to be easy to form, bend, and break.



THERMOPLASTIC



THERMOSET

Polymer Applications

Applications and **Examples**

- Adhesives and glues
- Containers
- Moldable products (computer casings, telephone handsets, disposable razors)
- Clothing and upholstery material (vinyls, polyesters, nylon)
- Water-resistant coatings (latex)
- Biodegradable products (corn-starch packing “peanuts”)
- Biomaterials (organic/inorganic intefaces)
- Liquid crystals
- Low-friction materials (teflon)
- Synthetic oils and greases
- Gaskets and O-rings (rubber)
- Soaps and surfactants

Introduction Cont….

- Some Polymeric products



Introduction Cont….

Composite Materials

- Made by combining two or more materials – often ones that have very different properties
- The two materials work together to give the composite unique properties

- Examples:

- **Natural**: Wood (Cellulose and lignin), bone (hydroxyapatite and collagen)
- **Synthetic:**
- Concrete
- Fiberglass; GFRP (Glass fibers embedded in polymer, e.g. epoxy)

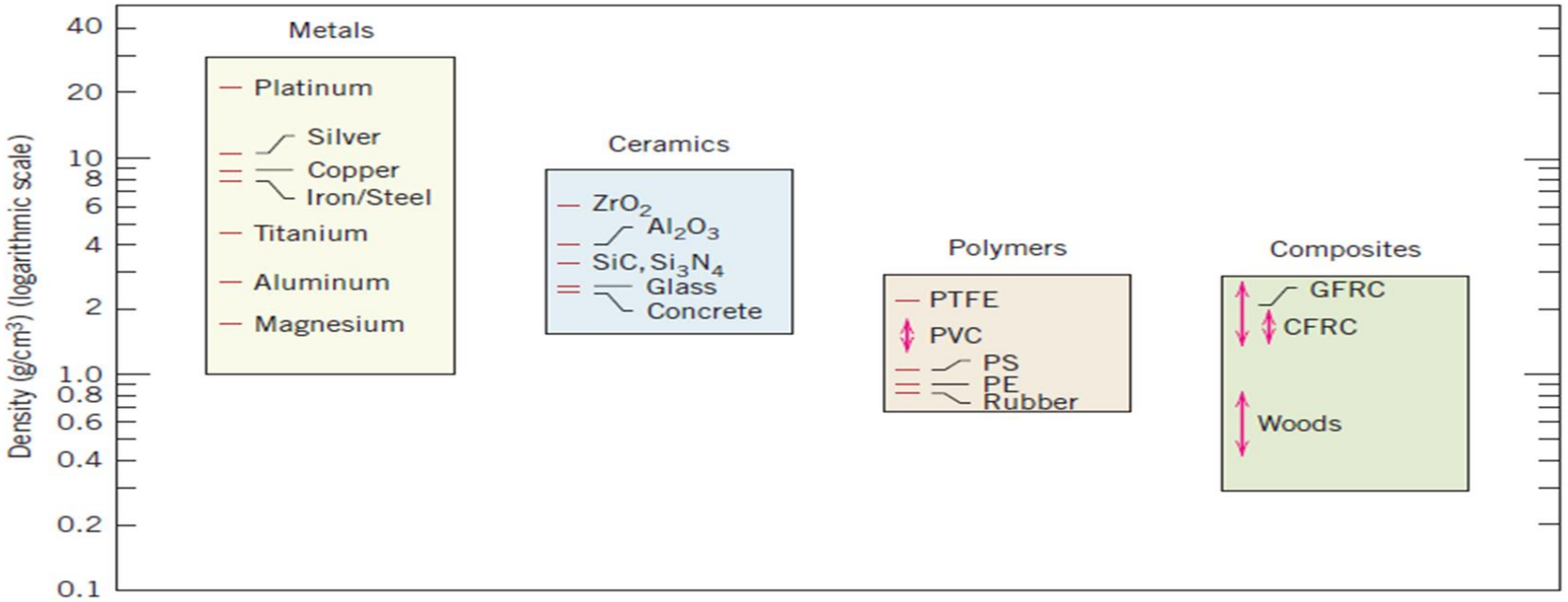
Properties: Strong, stiff-fibers, flexible and light-polymer

- Carbon fiber-reinforced polymer (CFRP) composite

CFRP has very peculiar application but very
expensive

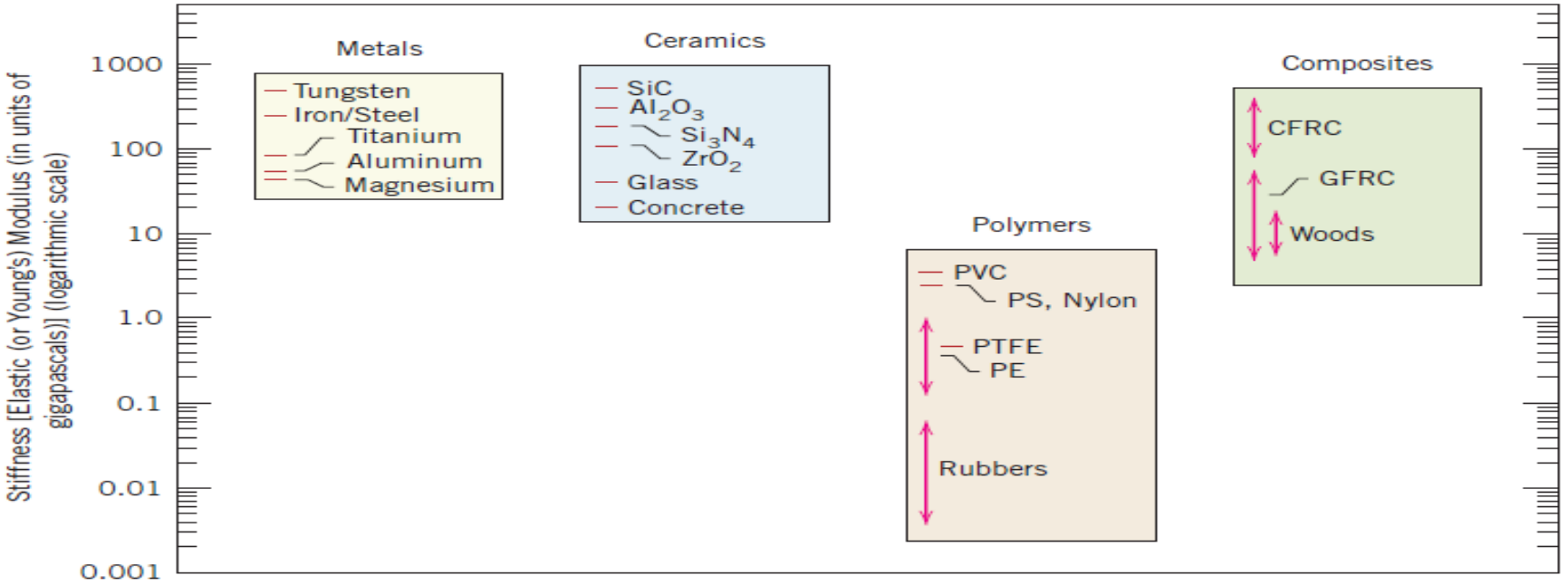
- **Aircraft and Aerospace, golf clubs, tennis rackets etc**

CLASSIFICATION OF MATERIALS



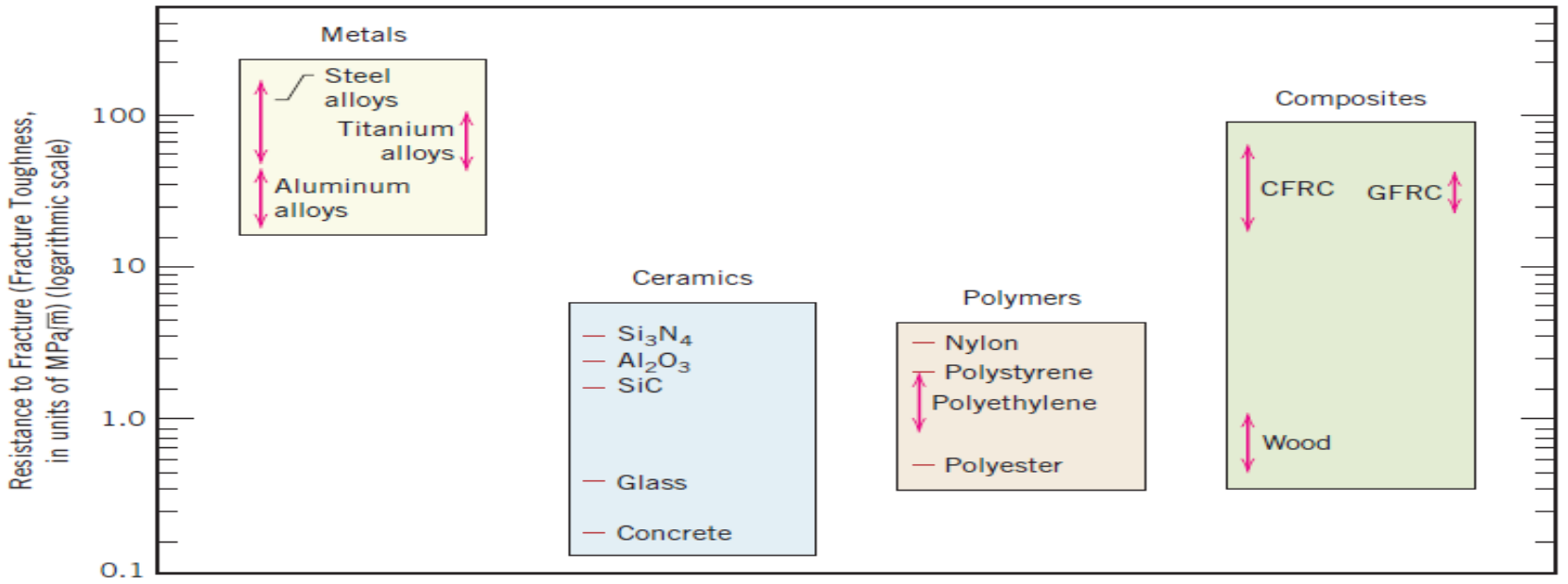
Bar-chart of room temperature density values for metals, ceramics, polymer and composite materials

CLASSIFICATION OF MATERIALS



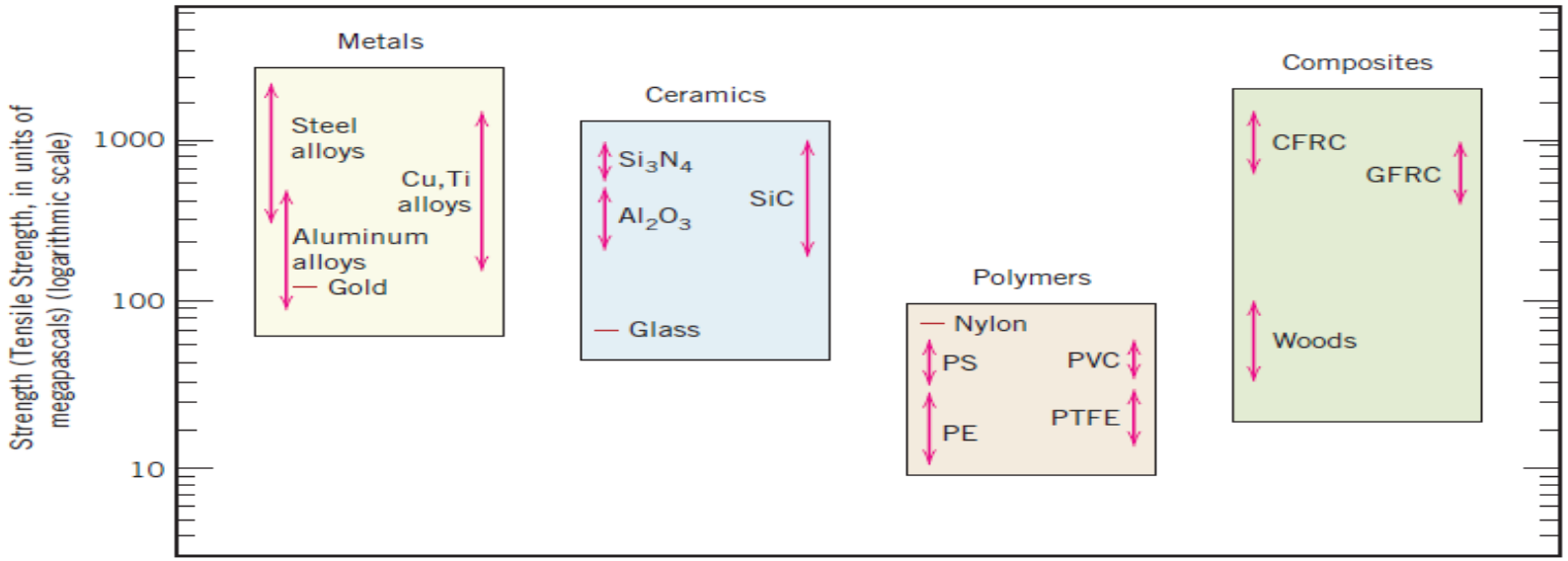
- *Bar-chart of room temperature stiffness(elastic modulus) values for metals, ceramics, polymer and composite materials*

CLASSIFICATION OF MATERIALS



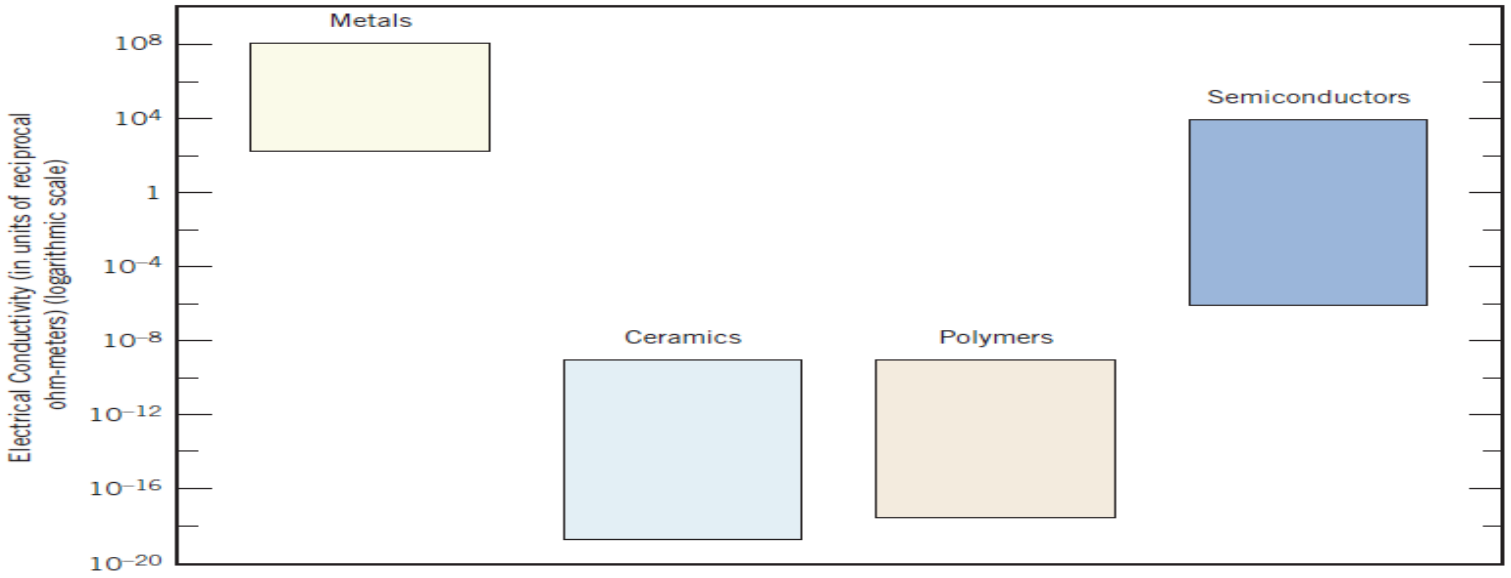
- *Bar-chart of room temperature resistance to fracture for metals, ceramics, polymer and composite materials*

CLASSIFICATION OF MATERIALS



- *Bar-chart of room temperature tensile strength for metals, ceramics, polymer and composite materials*

CLASSIFICATION OF MATERIALS



- *Bar-chart of room temperature electrical conductivity ranges for metals, ceramics, polymer and composite materials*

Assignment due 7th July 2022

1. Write on

✓electron quantum numbers

✓Periodic table

2. What are the different types of bonds found in solid?

✓Describe how bonding energies and forces affect the properties of solid materials

CRYSTAL STRUCTURE OF SOLIDS

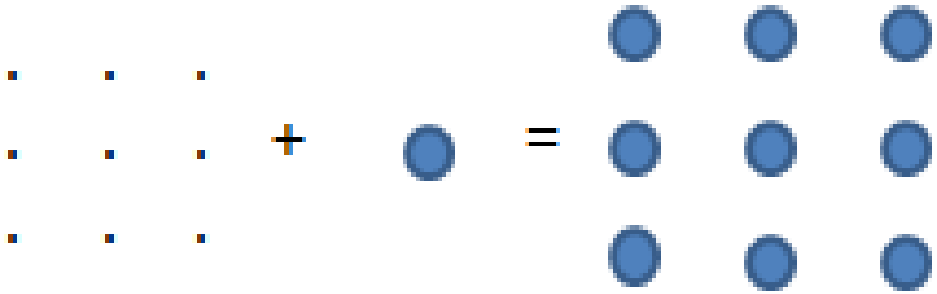
Structures of Materials

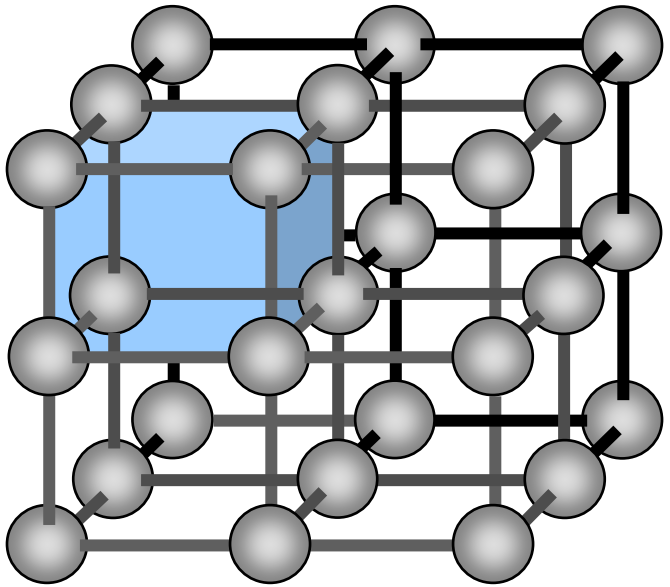
ISSUES TO ADDRESS...

- How do atoms assemble into solid structures?
- How do the structures of metals differ from those of ceramic materials?
  - How does the density of a material depend on its structure?
  - How does the mechanical properties of a material depend on its structure?

THE STRUCTURE OF CRYSTALLINE SOLIDS

The Crystal Lattice





- Lattice -3 dimensional array of points coinciding with atomic positions (sphere centers)

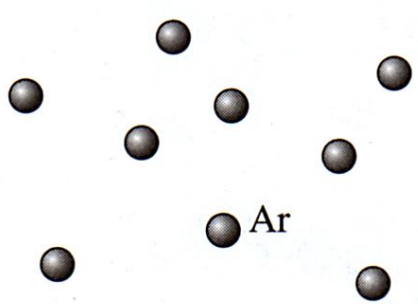
Point lattice + atom/atom group (Basis) = Crystal

A point lattice is made up of regular, repeating points in space. An atom or group of atoms are tied to each lattice point

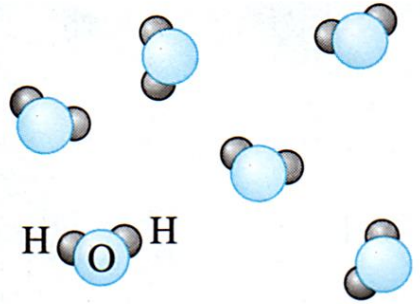
THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystal Structures

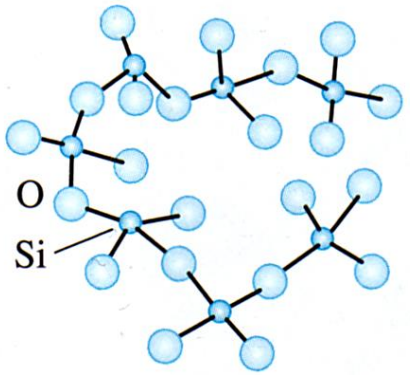
- Solid materials may be classified according to the regularity with which atoms or ions are arranged with respect to one another.
- A crystalline material is one in which th ting or periodic array over large atomic dis ion, the atoms will position themselves in a pattern, in which each atom is bonded
- Levels of atomic arrangement in mater
  - No order
  - Short-range order
  - Long-range order.



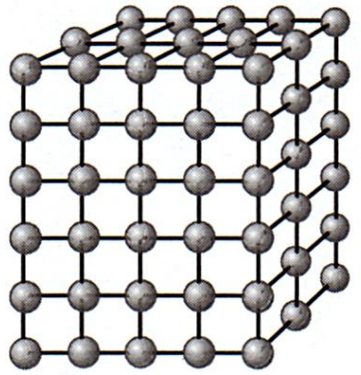
(a)



(b)



(c)


(d)

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystal Structures

Fundamentals

- Crystalline solids exhibit long-range order atomic arrangement
- Upon solidification, atoms position themselves in a repetitive three dimensional pattern in which each atom is bonded to its nearest-neighbour atoms
- On solidification, all metals, most ceramic and a few polymeric materials form crystal structures.
- In the absence of long range order, we have non-crystalline or amorphous materials.
- Some properties of crystalline solids depend on the type of crystal structure it has.
- Crystal structure: is the manner in which atoms, ions or molecules of a solid are spatially arranged.

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystal Structures

Fundamentals

- ***Atomic hard sphere model***
  - When describing crystals structures, atoms/ions are assumed as hard spheres having well-defined diameters
  - Spheres representing nearest neighbors touch each other

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystal Structures

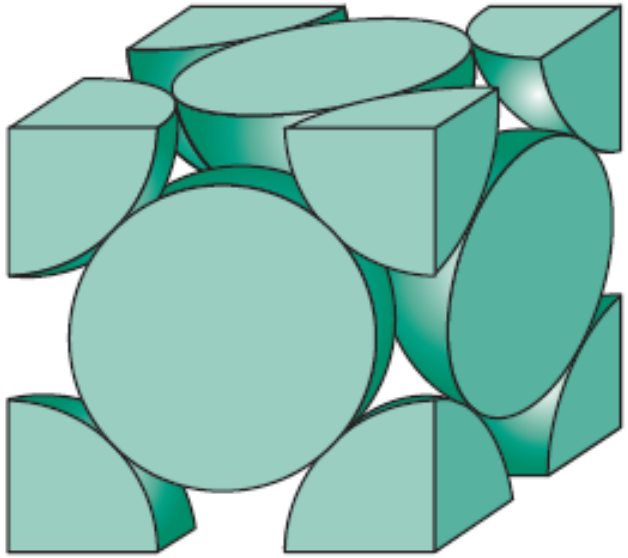
Unit Cells

- A sub-structure of repeat entities that is used to represent the crystal structure
- Building blocks of a crystal structure assuming the hard sphere model
- Mostly parallelepipeds or prisms with 3 sets of parallel faces whose corners coincide with atomic centers for convenience
- Shows a symmetry of the crystal with atom positions generated by translation along it's edges

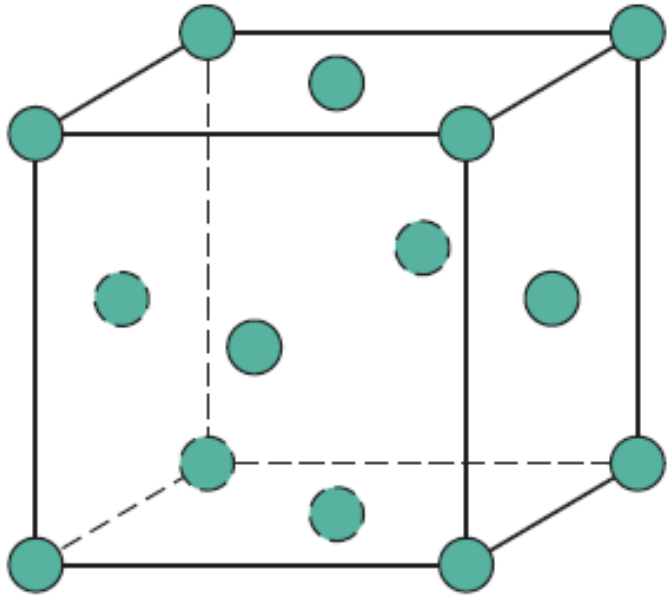
THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystal Structures

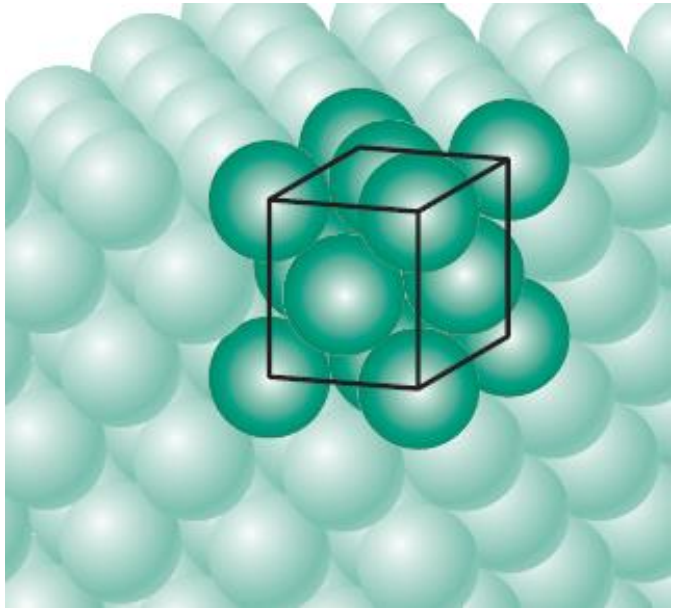
Fundamentals



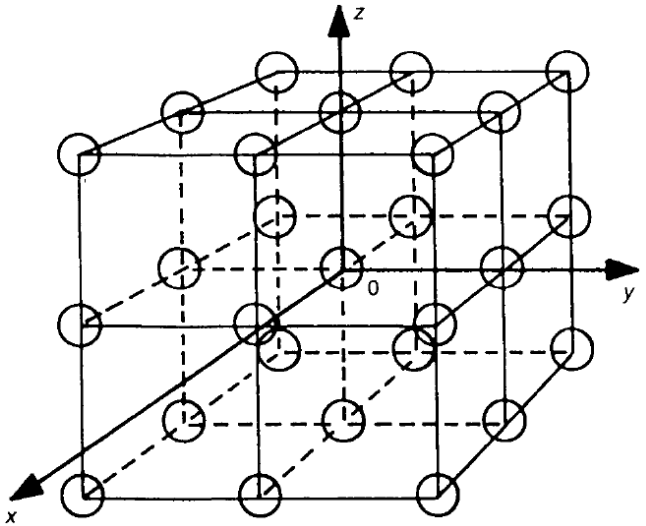
Hard sphere unit cell representation (FCC)



Reduced-sphere unit cell (FCC)



Aggregate of many atoms (FCC)



Eight unit cells put together (SC)

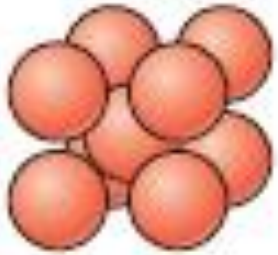
THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystal Structures

Fundamentals



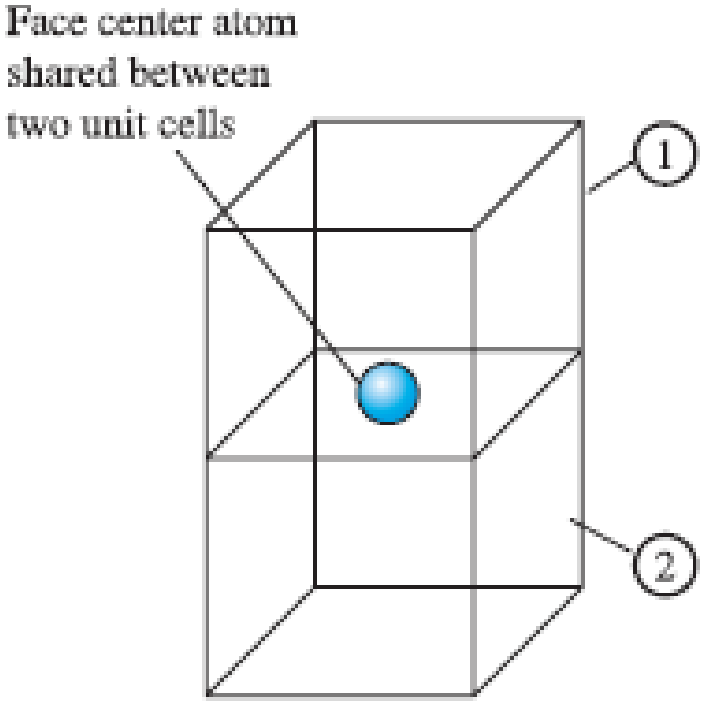
Simple cubic

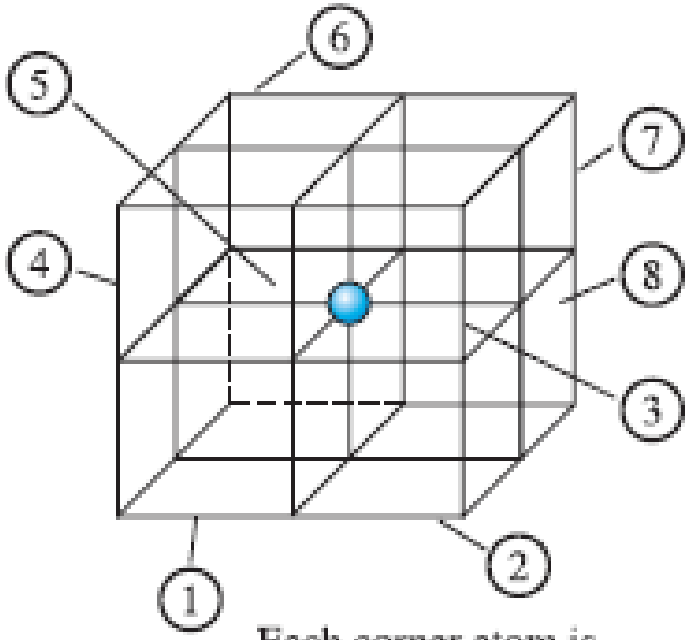


Body-centered cubic



Face-centered cubic



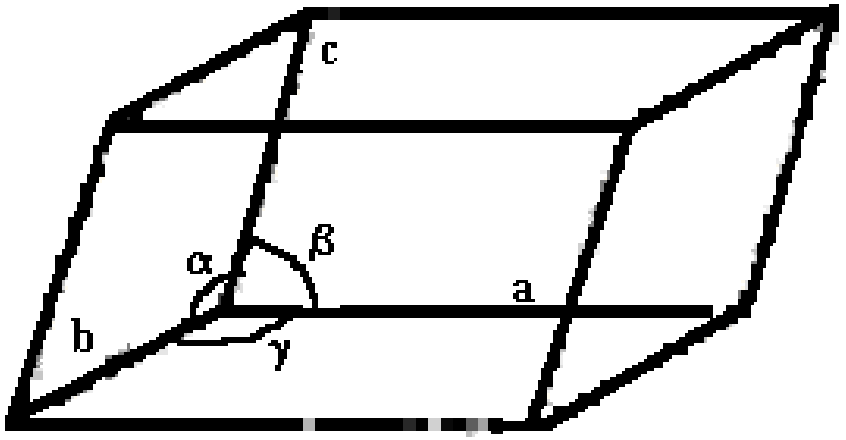


Each corner atom is shared by 8 unit cells (1–4 in front, 5–8 in back)

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystal Systems

- There are so many possible crystal structures
- For convenience, we mostly group structures according to unit cell configurations and/or atomic arrangements
- One method is by considering the unit cell geometry only
- 7 possible combinations is possible each being a distinct crystal system
- x, y, z coordinate system is used with origin at one corner of the parallelepiped unit cell corner
- Six parameters are used here to define a partic *ice parameters*
  - ***a***, ***b*** **and** ***c*** **i.e. edge lengths and**
  - ***α***, ***β***, ***γ*** **i.e. 3 interaxial angles**



THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystal Structures

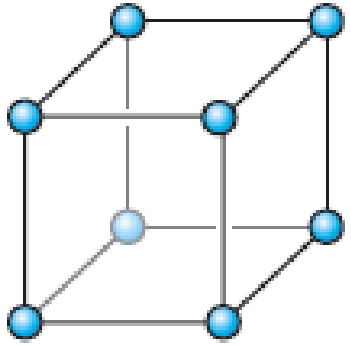
Crystal system	Axial relationship	Interaxial angles
Cubic	a = b= c	$\alpha = \beta = \gamma = 90°$
Tetragonal	a = b ≠ c	$\alpha = \beta = \gamma = 90°$
Hexagonal	a = b ≠ c	$\alpha = \beta = 90°, \gamma = 120$
Orthorhombic	a ≠ b ≠ c	$\alpha = \beta = \gamma = 90°$
Rhombohedral (trigonal)	a = b= c	$\alpha = \beta = \gamma \neq 90°$
Monoclinic	a ≠ b ≠ c	$\alpha = \gamma = 90°, \neq \beta$
Triclinic	a ≠ b ≠ c	$\alpha \neq \beta \neq \gamma \neq 90°$

THE STRUCTURE OF CRYSTALLINE SOLIDS

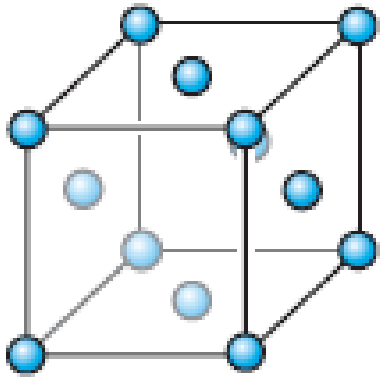
Crystal Structures

Crystal Systems

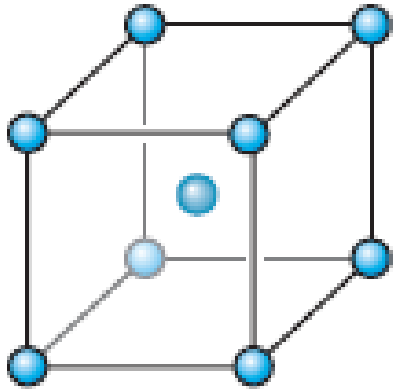
- Cubic system has 3



Simple Cubic

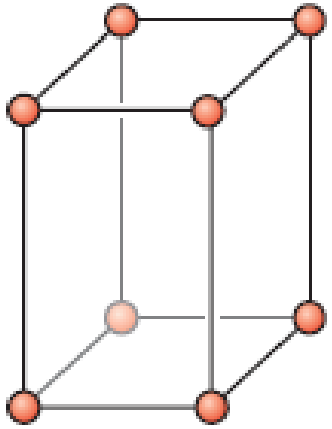


Face Centred Cubic

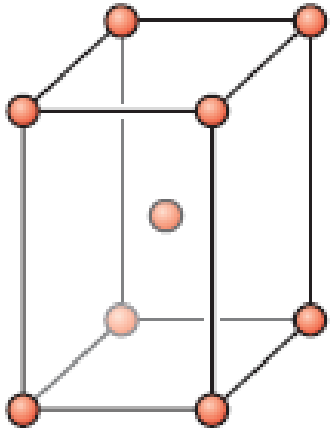


Body Centred Cubic

- Tetragonal system has 2 structures



Simple Tetragonal



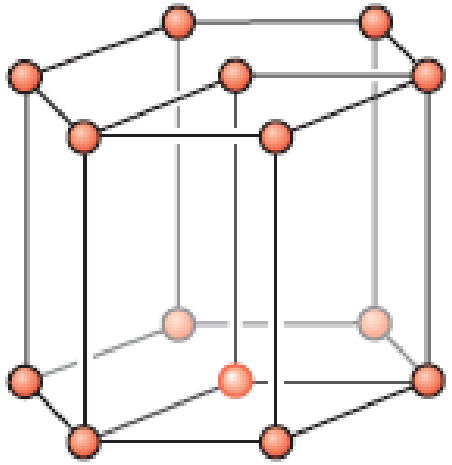
Body Centred Tetragonal

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystal Structures

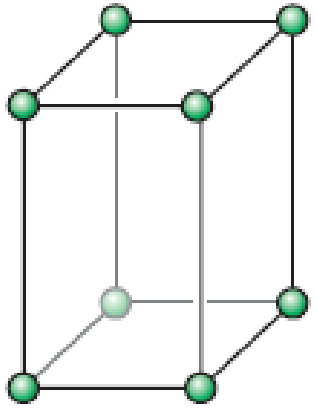
Crystal Systems

- Hexagonal system has just 1 structure

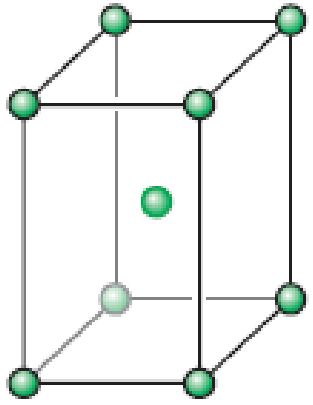


Hexagonal

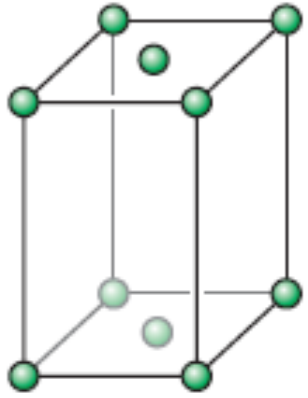
- Orthorhombic system has 4 structures



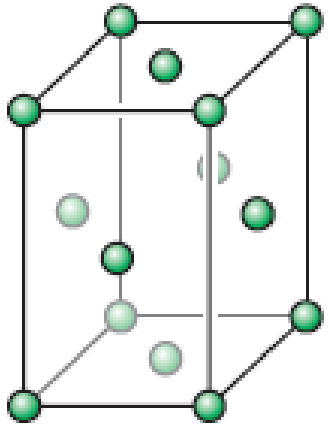
Simple Orthorhombic



Body-Centred Orthorhombic



Base-Centred Orthorhombic



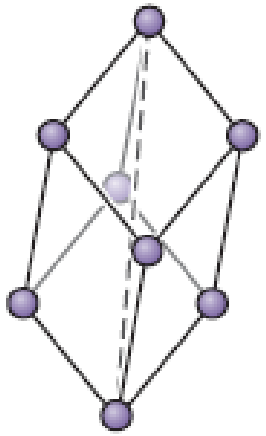
Face Centred Orthorhombic

THE STRUCTURE OF CRYSTALLINE SOLIDS

Crystal Structures

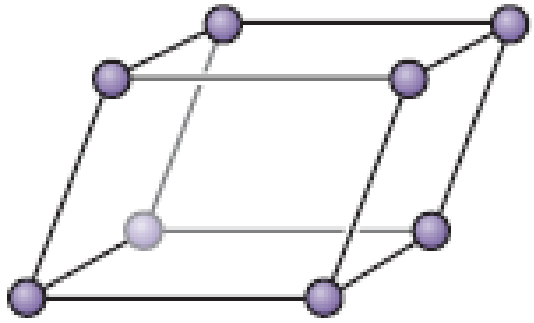
Crystal Systems

- Rhombohedral (trigonal) system has 1 structure



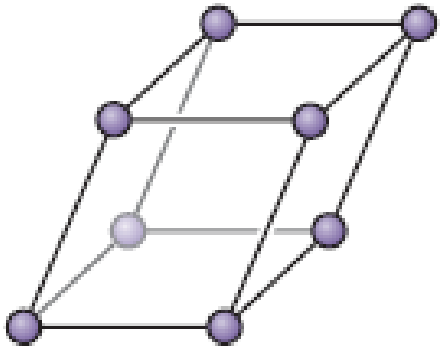
Rhombohedral

- Triclinic system has 1 structure

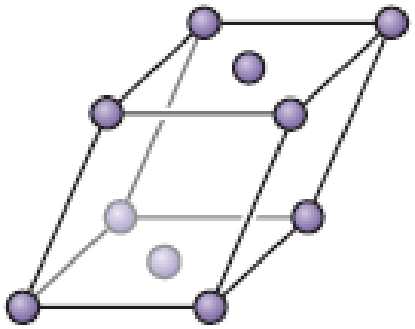


Triclinic

- Monoclinic system has 2 structures

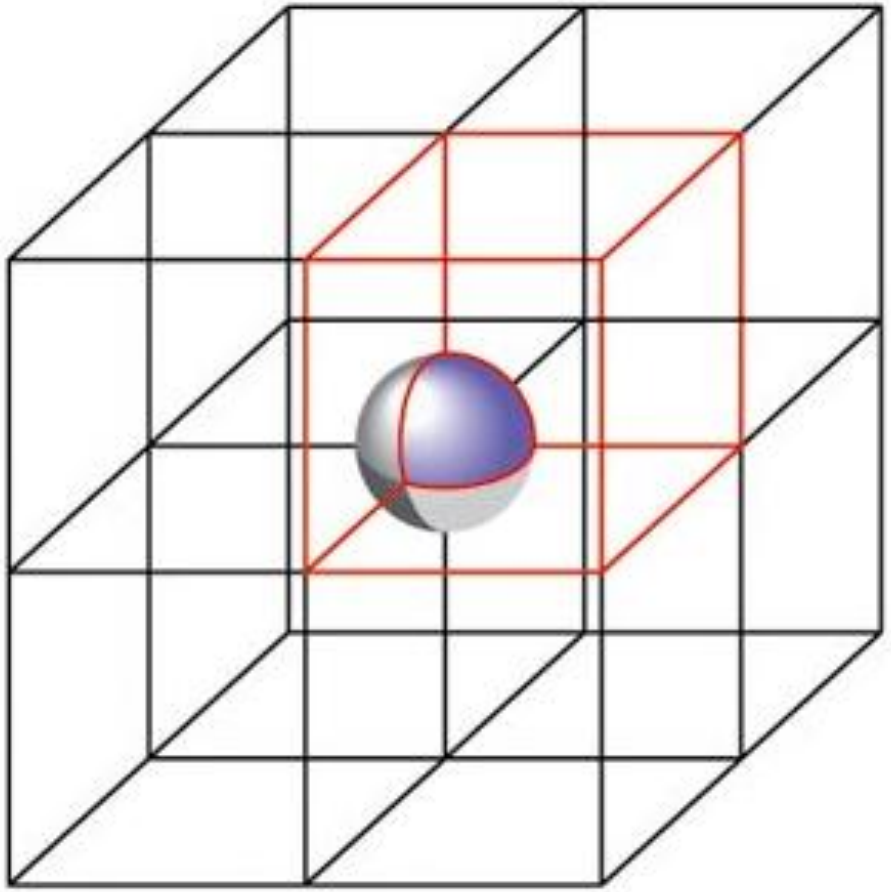


Simple Monoclinic



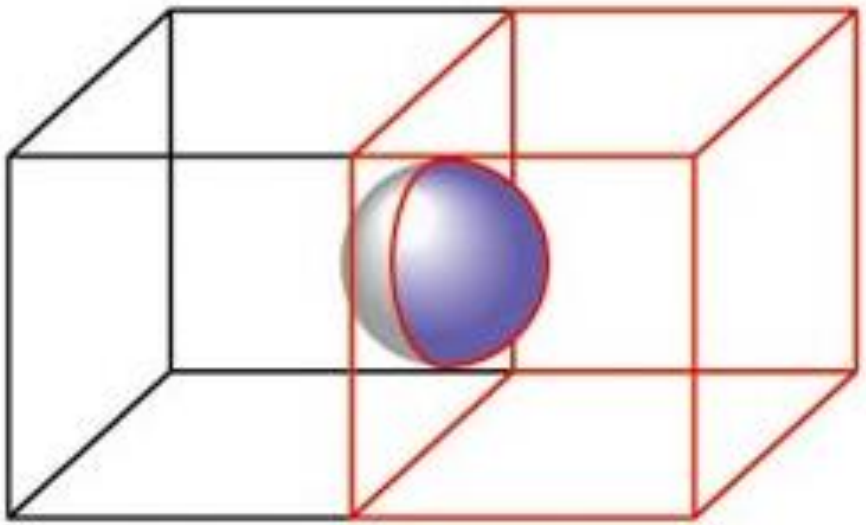
Base-Centred Monoclinc

of Atoms/Unit Cell



- For atoms in a cubic unit cell:
  - Atoms in corners are $\frac{1}{8}$ within the cell

of Atoms/Unit Cell



- For atoms in a cubic unit cell:
  - Atoms on faces are $\frac{1}{2}$ within the cell

of Atoms/Unit Cell

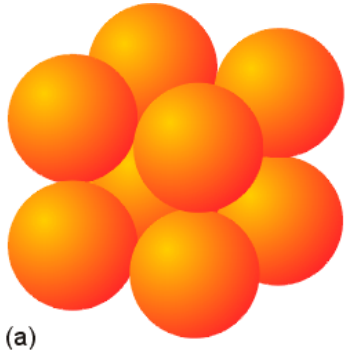
For atoms in a cubic unit cell:

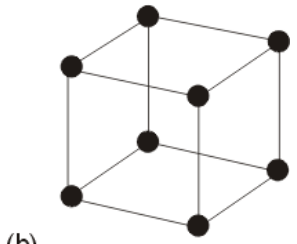
Atom Location	Number of Unit Cells Sharing Atom	Fraction of Atom Within Unit Cell
Corner	8	1/8 or 12.5%
Edge	4	1/4 or 25%
Face	2	1/2 or 50%
Anywhere else	1	1 or 100%

*It is only the position of the center of the atom that matters. Atoms that reside near the boundary of the unit cell but not on a corner, edge, or face are counted as residing 100% within the unit cell.

Simple Cubic Structure (SC)

- Rare due to poor packing (only Po has this structure)
- The atoms occupy the corners of a cube.
- Close-packed directions are cube edges.
- The coordination number is 6, and the packing efficiency is only 52.4%





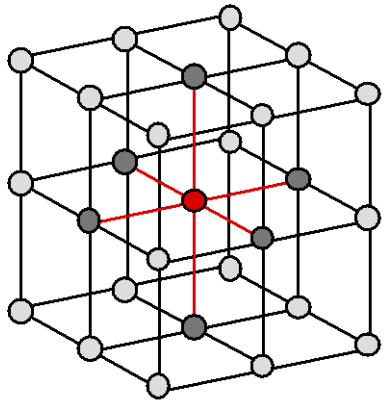


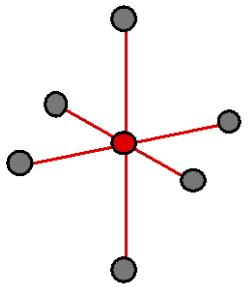
(Courtesy P.M. Anderson)



(a) Primitive cubic metal
1 atom per unit cell

- Coordination # = 6
  (# nearest neighbors)



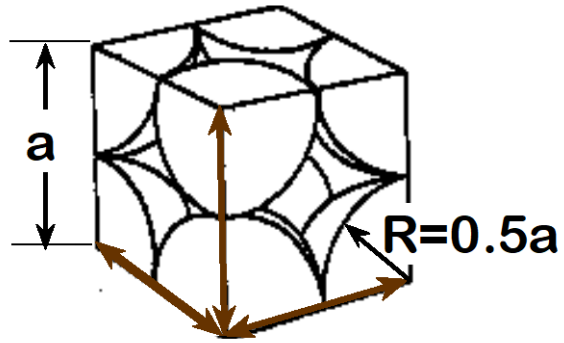


Atomic Packing Factor: SC

$$APF = \frac{\text{Volume of atoms in unit cell}^*}{\text{Volume of unit cell}}$$

*assume hard spheres

- APF for a simple cubic structure = 0.52



close-packed directions

contains 8 x 1/8 =

1 atom/unit cell

$$APF = \frac{\frac{\text{atoms}}{\text{unit cell}} \cdot 1 \cdot \frac{4}{3}\pi (0.5a)^3 \cdot \frac{\text{volume}}{\text{atom}}}{a^3 \cdot \frac{\text{volume}}{\text{unit cell}}}$$

Adapted from Fig. 3.19, *Callister 6e.*